Terrain Perceptor

Team Name: NeuralNomads
Hackathon: TechnoMania 2.0 – Duality AI Integrated Autonomy Challenge
Elective Track: NLP – Scene Reasoning
Project Tagline: *“From Desert Images to Human-Readable Terrain Insights”*

1. Methodology

Objective:
Train a semantic segmentation model on synthetic desert environments and implement an NLP module to generate human-readable situational summaries based on segmentation outputs.

Workflow Overview:

1. **Data Preparation:**
   - Dataset: Synthetic desert images from FalconCloud (Train, Validation, Test sets)
   - Classes included: Background, Trees, Lush Bushes, Dry Grass, Dry Bushes, Ground Clutter, Logs, Rocks, Landscape, Sky
2. **Foundation Model Training:**
   - Training environment: Miniconda “EDU” environment
   - Scripts used: train.py, test.py
   - Data augmentation: Random rotations, flips, brightness variations to simulate environmental diversity
3. **Elective Track – NLP Module:**
   - Script: nlp_reasoner.py
   - Input: Segmentation mask metadata (percentage of pixels per class)
   - Output: Human-readable situational reports for UGV operators
   - Example:
     - Input: Rocks occupy 23% of frame, Dry Grass 16%
     - Output: *“High hazard density detected; Rocks occupy ~23% of the frame. Proceeding with caution.”*
4. **Evaluation Metrics:**
   - IoU (Intersection over Union) for segmentation
   - Mean IoU across all classes
   - Per-class IoU for detailed insights

GitHub Repository: https://github.com/lucifer20004/terrain_perceptor

2. Results & Performance Metrics

Overall Model Performance:

- **Mean IoU:** 0.2314

Per-Class IoU:

Class	IoU Score
Background	0.0000
Trees	0.0768
Lush Bushes	0.0066
Dry Grass	0.1615
Dry Bushes	0.0660
Ground Clutter	0.0603
Logs	0.0061
Rocks	0.2293
Landscape	0.5350
Sky	0.9388

Analysis:

- **Strong performance** on large, visually distinct classes: Sky (0.9388 IoU), Landscape (0.5350 IoU)
- **Weak performance** on small or occluded objects: Logs (0.0061 IoU), Lush Bushes (0.0066 IoU), Background (0.0000 IoU)
- Moderate detection of Rocks (0.2293 IoU) and Dry Grass (0.1615 IoU)

NLP Module Performance:

- Despite low IoU for some classes, the NLP module effectively **highlights hazards and critical terrain features**

Key Takeaways:

1. Foundation model identifies major environmental classes reliably
2. Smaller and occluded objects remain challenging
3. NLP reasoning is robust enough to provide **usable situational insights**

4. Challenges & Solutions

Challenge	**Solution**
Small-object detection (Logs, Lush Bushes)	Applied data augmentation with occlusion and sparse object simulation
Class imbalance (Sky and Landscape dominate pixel counts)	Weighted loss function during training to emphasize minor classes
Generating interpretable NLP summaries	Designed threshold-based rules on class densities for hazard assessment
Model inference speed	Optimized batch size and reduced model complexity for real-time usability

4. Conclusion & Future Work

Conclusion:

- Successfully trained a semantic segmentation model on synthetic desert data
- Developed an NLP module (**Terrain Perceptor**) that generates **human-readable terrain summaries**
- Provided actionable situational awareness for off-road UGVs

Future Improvements:

1. **Small-object recognition**: Integrate attention mechanisms or higher-resolution inputs to improve IoU on Logs and Lush Bushes
2. **Domain robustness**: Use generative AI to synthesize rare edge-case scenarios like sandstorms or nighttime conditions
3. **Integration with voice interface**: Combine with Speech-to-Text track to allow **voice-activated terrain queries**
4. **Self-supervised learning**: Reduce dependency on labeled data while improving generalization

5. References & Resources

- Dataset: FalconCloud Synthetic Desert Environment
- Falcon Platform: Create Account – FalconCloud by Duality AI
- GitHub Repository: Terrain Perceptor
- Scripts: train.py, test.py, nlp_reasoner.py
- UI Screenshot: